

Jonghyun Shin

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About

I am a second-year M.S.-Ph.D. integrated student in Artificial Intelligence at **Korea University**, advised by **Prof. Sejun Park**. My research studies how finite numerical precision and finite randomness affect the theoretical behavior of machine learning algorithms, with particular interests in **generalization**, **algorithmic stability**, and **differential privacy**.

Education

Korea University M.S.-Ph.D. Integrated Program in Artificial Intelligence • GPA: 4.5/4.5; advised by Prof. Sejun Park.	Seoul, South Korea Sep. 2025 – Present
Korea University B.S. in Mathematics Education / Statistics (Double Major) • GPA: 4.31/4.5.	Seoul, South Korea Mar. 2019 – Aug. 2025
Gyeongnam Science High School (GSHS)	Gyeongsangnam-do, South Korea Mar. 2017 – Dec. 2019

Publications & Preprints

International Conference

[C1] Jonghyun Shin, Namjun Kim, Geonho Hwang, Sejun Park, Minimum Width for Universal Approximation using Squashable Activation Functions, Proceedings of the International Conference on Machine Learning (ICML), 2025 [arXiv]

Preprint

[P1] Jonghyun Shin, Sejun Park, Uniform Stability and Generalization Error of GD and SGD on Fixed-Point Parameters, [arXiv]

Research Experience

Efficient Inference and Learning Lab Undergraduate Research Student • Advised by Prof. Sejun Park. • Research topic: characterizing minimum width for universal approximation using general activation functions.	Korea University, Seoul Dec. 2023 – Aug. 2025
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Awards & Honors

Master's Excellence Scholarship in Science and Engineering Korea Student Aid Foundation	Sep. 2025 – Aug. 2027
Dean's List Korea University	Mar. 2021, Sep. 2023

Teaching & Academic Service

Teaching Assistant, Korea University Discrete Mathematics (Spring 2025); Calculus (Fall 2025)	2025
Teaching Practicum, Daeshin High School Seoul, South Korea	Apr. 2025
Reviewer, NeurIPS	2026

Skills

Languages: Korean (native), English | **Programming:** Python, R | **Tools:** $\LaTeX$